

Vectorial vs. Alignment-Based Models of Locust Collective Motion

Asher Albrecht and George Stephenson

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1. Problem and Significance

Desert locust (*Schistocerca gregaria*) swarms are among the most dramatic and consequential examples of collective motion in nature. During the 2019 – 2022 East African outbreak swarms reached 2,400 km² in size and over 20 million people faced severe food insecurity. Understanding the local interaction rules that give rise to coordinated swarm movement is therefore not only a fundamental question in collective behavior but one with direct implications for prediction and management of locust outbreaks.

For three decades, self-propelled particle (SPP) models (most notably the 1995 Vicsek model) have been the dominant framework for collective motion. In these models, agents align with the average heading of their neighbors, producing a density-driven phase transition from disorder to order. Buhl et al. (2006) reported that desert locusts in ring arenas exhibit exactly this transition, seemingly validating SPP models for biological swarms.

Sayin et al. (2025) challenge this interpretation. Through field experiments, virtual reality studies, and computational modeling, they show that locusts do not explicitly align with neighbors' headings. Instead, individuals steer toward the *positions* of conspecifics ahead of them – a “pull” mechanism. These two rules make different predictions at the group level, but no one has yet tested whether the pull model better reproduces real field data at scale.

We will implement both a Vicsek alignment model and a simplified pull model as agent-based simulations, then quantitatively compare their outputs against ~20,000 real locust trajectories from the Weinburd et al. (2024) Dryad dataset.

2. Background and Related Work

The Vicsek model (1995) is the canonical SPP model: agents move at constant speed and update their heading to match the local average, plus noise. Couzin et al. (2002) extended this with concentric repulsion/orientation/attraction zones. Buhl et al. (2006) provided apparent empirical support for SPP predictions in locust nymphs. However, Sayin et al. (2025) identified a fundamental confound in the ring-arena setup: density and order are inherently correlated, making it impossible to determine whether density causes alignment or whether alignment emerges independently.

To disentangle these factors, Sayin et al. used three complementary approaches. First, sensory deprivation experiments during the 2020 East African outbreak established that vision, not olfaction or polarized light, is necessary and sufficient for social coordination. Second, a panoramic virtual reality system allowed independent manipulation of density and order for freely moving locusts, revealing that alignment depends on swarm order ($R^2 = 0.82$) but not density ($R^2 = 0.03$). Third, they ruled out the optomotor response (tracking global visual motion) as the mechanism, since solitary and gregarious locusts show identical optomotor responses yet only gregarious locusts align with virtual swarms. Instead, locusts appear to use a vectorial bearing-based “pull” toward individual conspecifics. Separately, Krongauz et al. (2024) modeled vision-based collective motion in locust-like agents with monocular perception. Weinburd et al. (2024)

published the first individual-level field trajectory dataset from hopper bands, containing over 3.3 million position records across four bands of Australian plague locusts.

3. Plan of Execution

Week 1 — Data and Literature. Download the Weinburd dataset and load it into Python. Compute baseline statistics from real trajectories: order parameter, speed distributions, neighbor density maps, and turning-angle distributions. Asher leads biological interpretation; George leads data pipeline. *Checkpoint:* We can reproduce basic trajectory statistics from the published data.

Week 2 — Model Implementation. Implement two models from scratch in Python. Model A (Vicsek): each agent aligns with neighbors' average heading within a radius, plus noise. Model B (Pull): each agent steers toward the position of nearest neighbors within a forward-facing cone ($\sim 30\text{--}45^\circ$ offset). Both share the same speed, interaction radius, and boundary conditions. Asher implements Vicsek; George implements pull. *Checkpoint:* Both models produce coherent collective motion independently.

Week 3 — Comparison. Match simulation parameters to the field data and run both models. Compare simulated vs. real statistics on four metrics: order parameter distributions, turning-angle distributions, neighbor density maps, and nearest-neighbor distances. *Checkpoint:* Quantitative comparison plots for all metrics.

Week 4 — Analysis and Writing. Interpret which model better matches the data and on which metrics. Discuss limitations (different locust species between Sayin et al. and the Weinburd dataset; simplified pull model vs. the full ring-attractor framework). Write the final paper and prepare the presentation. *If all goes well:* test a hybrid model blending both mechanisms.

4. Future Directions

Beyond this class, this work could be extended by implementing the full neural ring-attractor model from Sayin et al. (rather than our simplified pull approximation) and testing it against additional species and datasets. The broader question (whether vectorial pursuit mechanisms generalize to other collectively moving species such as fish schools or bird flocks) remains open and would be a natural next step.

5. Bibliography

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